

IDUG

2025



EMEA Db2 TECH CONFERENCE

Dusseldorf | October 26 - 30

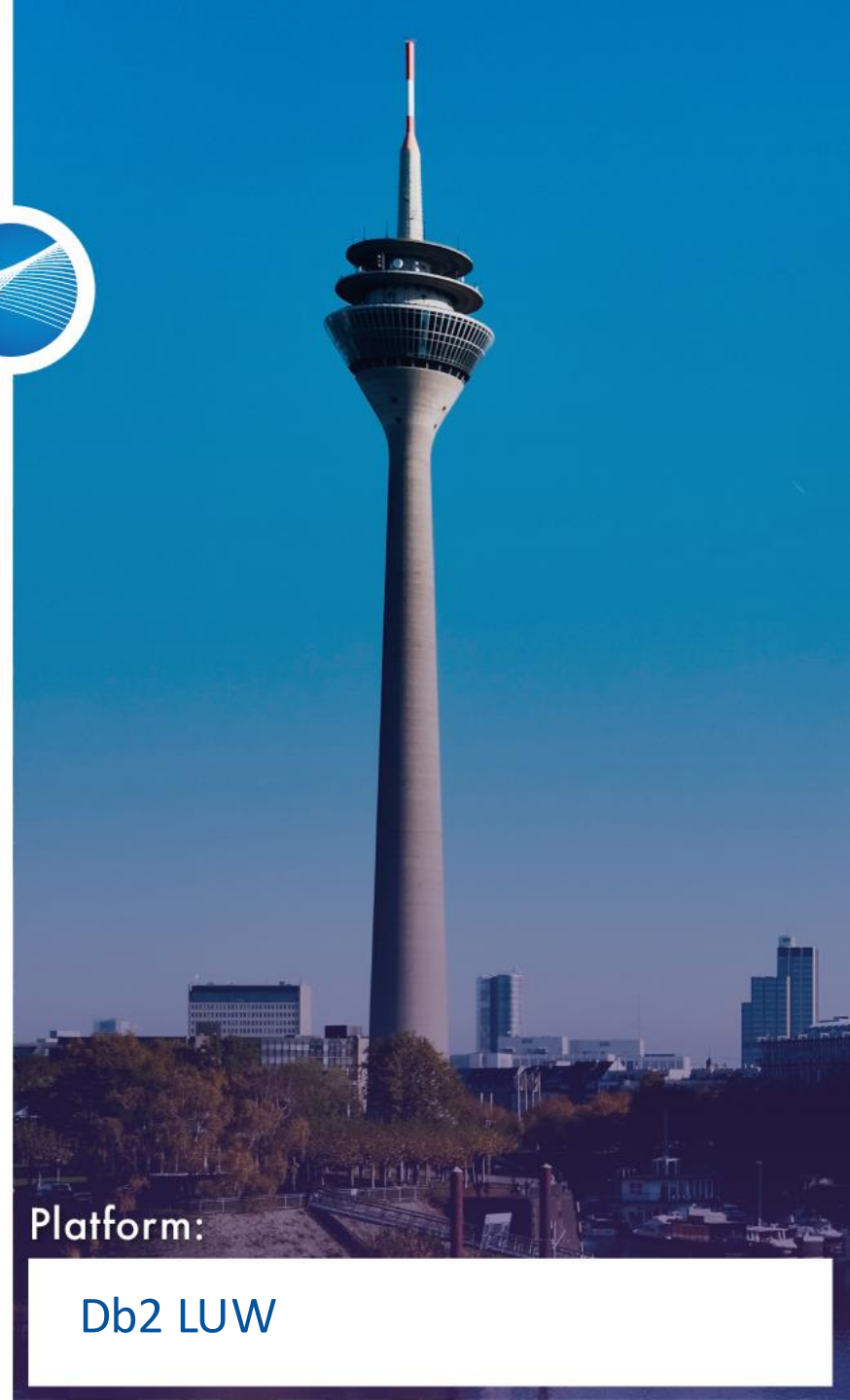
**Hands-On Workshop: Building LLM Applications
with Db2**

Shaikh Quader, IBM

Session Code: SESS-77

Platform:

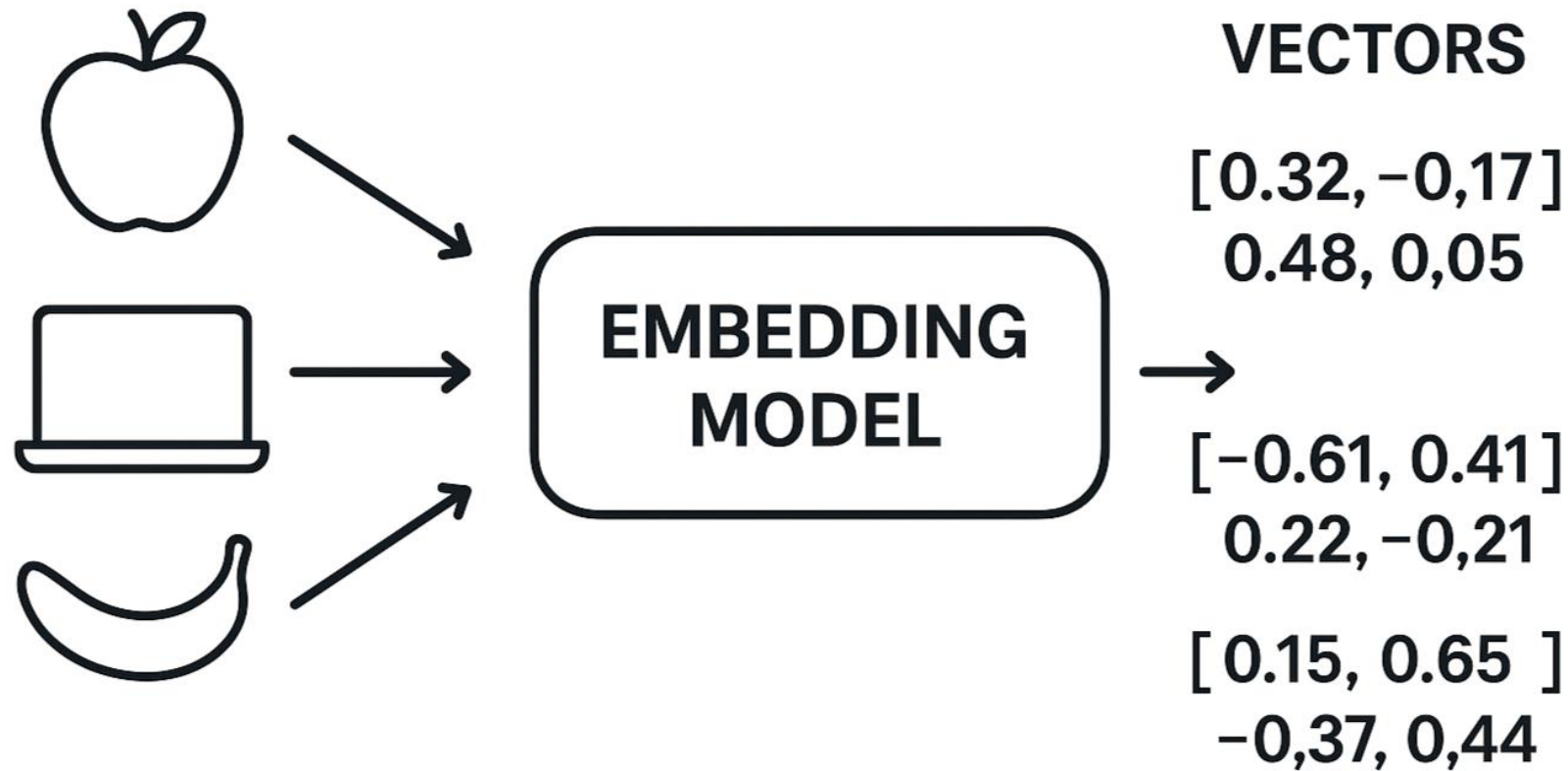
Db2 LUW



Which Two Are More Similar?

#IDUGDb2







So, what? What can Vectors do for you?



Find similar
songs you'll like



Shop by
taking a photo



Movies you
might enjoy



Find your
photos by faces



Search using
pictures



Ask questions,
get answers

VECTOR type:

- FLOAT32 vectors
- INT8 vectors
- Max dimension:
 - 8168 (FLOAT 32)
 - 32672 (INT8)

VECTOR Functions:

- VECTOR_DISTANCE – between vectors
- VECTOR constructor: string -> vector
- VECTOR_SERIALIZE – vector -> string
- *VECTOR_NORM* – vector magnitude
- VECTOR_DIMENSION_COUNT – number of dimensions

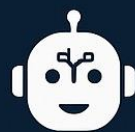
**LLM Apps Dev Support:
LangChain Connector**

Db2 LangChain Connector

IBM Db2



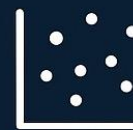
LangChain



AI Agent



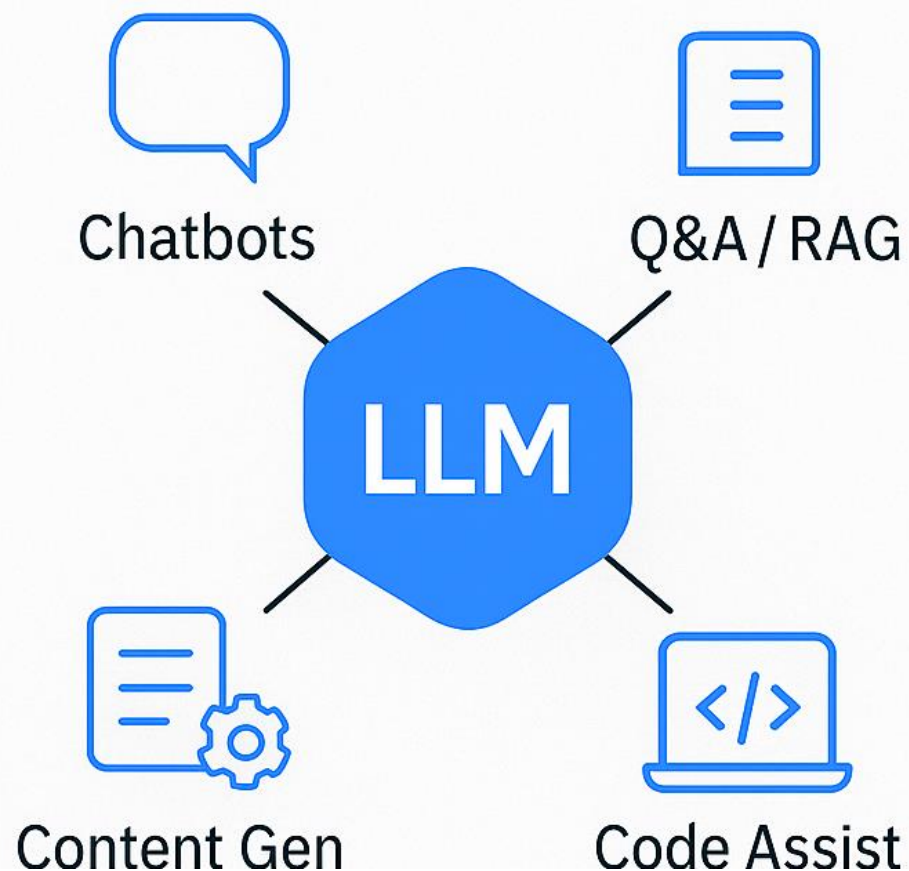
RAG

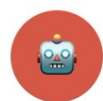


Vector
Search

Enabling modern AI use cases: RAG · AI Agents · Semantic Search

Apps that use **Large Language Models** to understand, generate, and interact with natural language.





Non-RAG LLM

Traditional Large Language Model

User Query

Direct input to LLM



Static Training Data

Fixed knowledge cutoff



Generation

Response based on training only

Knowledge-Cutoff / Static Knowledge, Hallucinations



RAG System

Retrieval-Augmented Generation

User Query

Input converted to vector



Retrieval

Search external knowledge base



Augmentation

Combine query + retrieved context

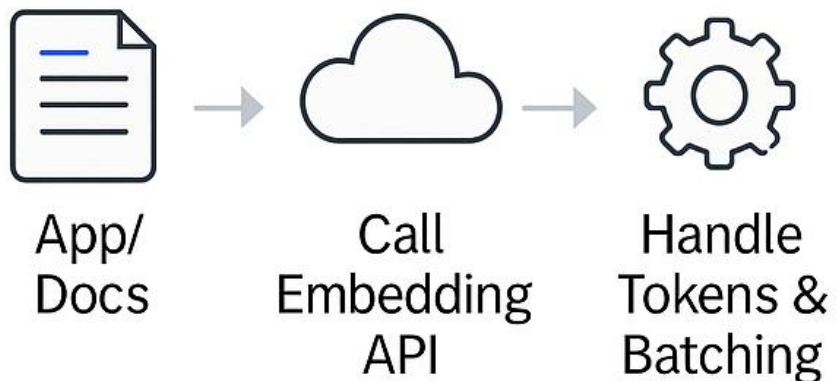


Generation

LLM generates grounded response

Dynamic Knowledge, Domain-Knowledge, Source citation

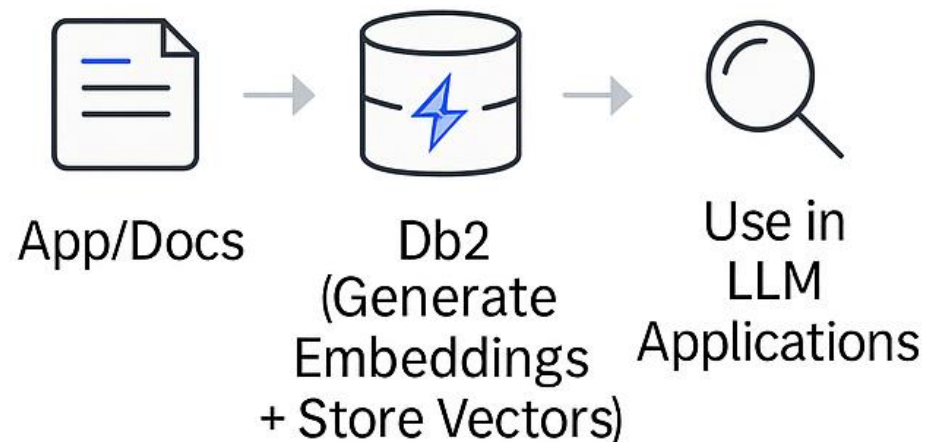
The Problem: Too Much Glue Code



Challenges:

Time • Fragility • Cost

The Solution SQL-Native Embeddings in Db2

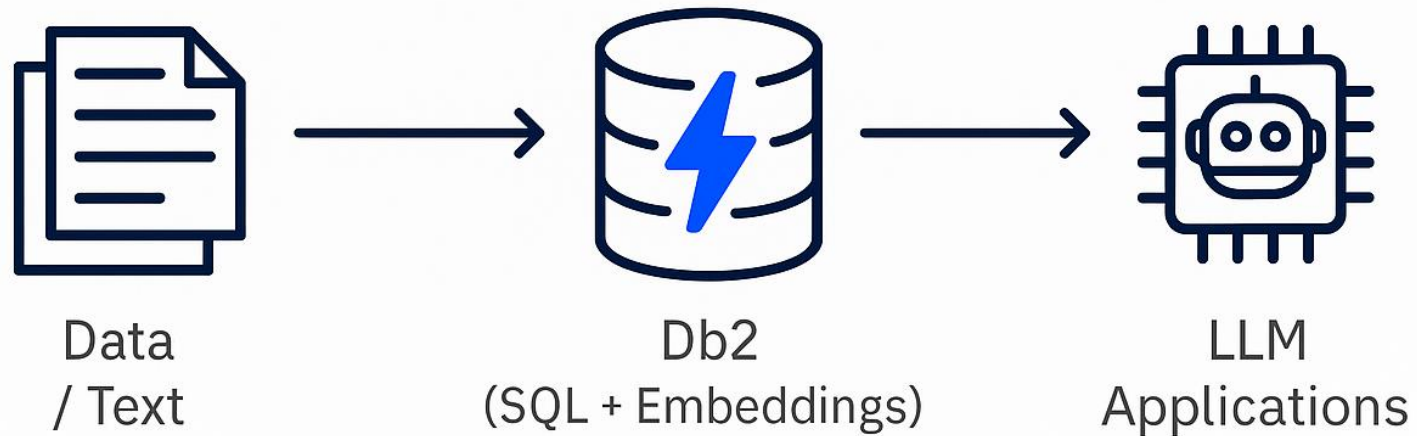


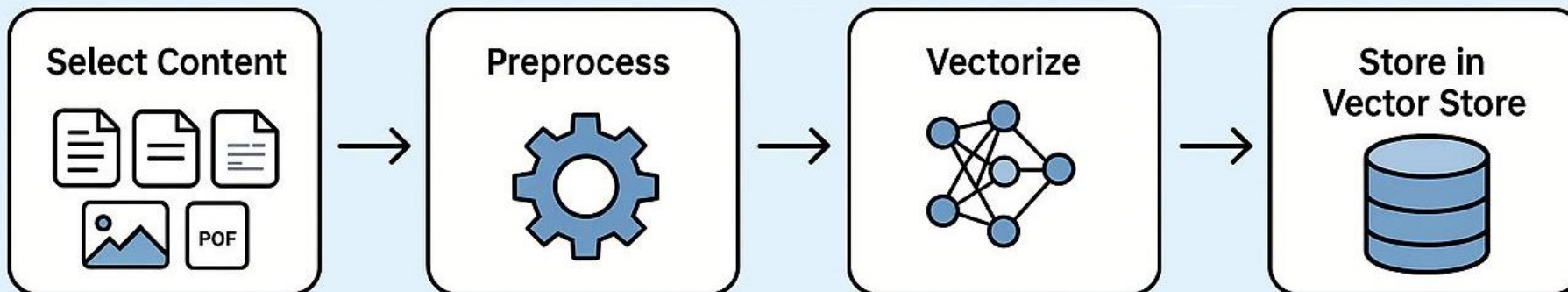
With Db2, register your embedding endpoint once. Then generate, store, and search embeddings SQL.

Demo

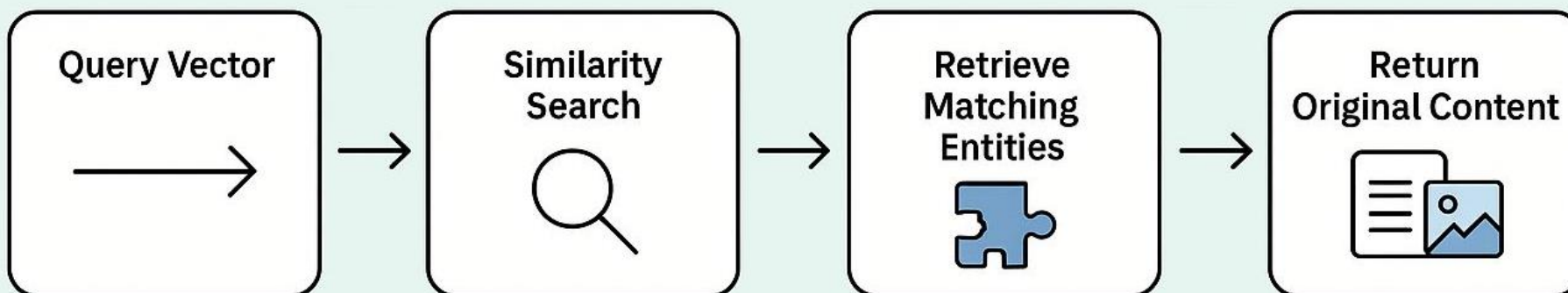
SQL-Native Embeddings in Db2

Simplifying LLM application development with fewer moving parts





PHASE 1: DATA INGESTION & VECTORIZATION



PHASE 2: QUERY & RETRIEVAL

ibm.biz/learndb2ai



Shaikh Quader

AI Architect @ IBM Db2 | Ph.D. Candidate in AI | IBM
Master Inventor | Sharing lessons from building & d...



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**Please fill out
your session
evaluation**

Platform:

Db2 LUW

An aerial photograph of Düsseldorf, Germany, featuring the Rhine river, the Esplanade promenade, and the skyline with the Esplanade Tower. The image is overlaid with a blue gradient.

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2025 EMEA **Db2** Tech Conference